Engineer to
Excel

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ENGINEERING

SIMATS

Saveetha Institute of Medical **And Technical** Sciences

(**Declared** as Bennes to be **University** under Section **3** of **UGC** Act 1966)

ASSESSMENT TOOL II- ASSIGNMENT

COURSE CODE: ITA14

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

C01: Apply the principle of the foundational concepts of ethical hacking, including types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

ASSIGNMENTS QUESTIONS

1. A company hires you to test its network but gives you only a one-line email authorization with no defined scope. Halfway through testing, you discover a critical vulnerability in a system that wasn't mentioned in the email. Do you exploit it to prove impact? Justify your answer using ethical hacking principles.
2. You are conducting a penetration test and discover that the client's production database contains real customer financial data. Your test plan calls for exploiting a SQL injection vulnerability. How do you proceed without causing harm, and what precautions should have been taken before the engagement began?
3. During a black-box penetration test, you accidentally trigger a denial-of-service condition on a live production server, causing an outage. Walk through what you should have done to prevent this and what your responsibility is now.
4. A client asks you to test their systems but insists you not inform their IT department, wanting to

test the team's real-world detection capability. What ethical and practical risks does this create, and how would you structure the engagement to protect yourself? 5. You're asked to perform a "red team" exercise versus a "vulnerability assessment." Using a real-world scenario of a bank wanting to test physical and digital security, explain how your approach, ethics, and rules of engagement would differ between the two.

Code of Conduct:

IP Venkata Sai Pradeep Reddy (192311364) (Name/Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Rubrics

Criteria	**Weightage**	**Level 4** **Exemplary**	**Level 3** **Proficient**	**Level 2** **Developing**	**Level 1** **Beginning**
Understanding of **Problem**	20	Clearly understands the problem and accurately	Good understanding with minor gaps	Basic understanding; some misinterpretation	Poor understanding or
Context					

requirements

of the problem

identifies all

unclear

data-handling

requirements

and

operations

Application of 30

Accurately

Applies

Applies basic Fails to apply

Concepts

applies

relevant

concepts with appropriate

appropriate

concepts

limited

concepts

Problem-

Solving

Approach

20

data structures correctly with accuracy

and

minor errors

algorithms to

solve

complex

problems

Logical, well- Mostly logical Basic approach Illogical structured,

or

approach with with some unclear approach

and efficient minor gaps

inconsistencies

approach

Accuracy

Solution

of 20

Completely Mostly correct Partially

Incorrect

or

correct code, with

output and errors

minor correct solution incomplete

solution

complexity

analysis with

proper

justification

Clarity

10

Clear, well- Understandable Limited clarity Poorly presented

organized, with

minor and

explanation

and properly clarity issues

and difficult to

understand

explained

solution

0 Un authorized

vulnerability Discovery.

If

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Approved

1) Avoid exploiting critical Venerability outside the Scope, I will exploiting vulnerability

vulnerability
without

2) Document
vulnerability

Permission

Safely without
causing damage

3) Inform the client immediately about
the

W obtain
written

4)

further tests.

5) Follow

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authorization before
perforraing

ethical principles such as
authorization

responsible disclosure &
avoiding thim

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proper permi

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- Cause Unexpected System

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Conclusi

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Controlled to evaluate

defection

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Capabilities without

Creating Unnecenary

risks.

5) Red Team

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Scenario:

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Assessmed",

test its physical

& digital

A Red

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2) Test's

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3) Requires

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3)

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4) stop testing

of

exposed

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5) Inform the

risks.

Sensitive information may

Client about the vulnerability & It's

Pre- cardion's before testing:

authorization & define the

1) Obtain

written

testing

ecope

2)

Arrange

backups

fest

accocuit's.

3) Establish data-protection

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rules

&

confidenti

9)

preaper

an

incident

response

& recovery

Without exposing plan.

Conclusion:

Tetting must be Conducted safely

Customer information or

disfruepting

businen

operation

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Accident

Denial of

Service

Outage !

Preventive

meaurel

1 Noord

Aggressive

testing against

live moduction

2) Establish request limet's & sade testing methods.

3) Define emersenly

stop

conditions before starting

4) Obtain approval for potentially disseuptive test's.

Responsibilities after

1 Stop Outage;

testing immediately

· stop

testing 2)

Notify client

8) Assist

{ responsible technical

team.

with system

of

Services.

4) Perserve

recovery &

restoration

Relevant logs &

incident

honestly

5) Indentify the

Measures.

Conclusion's

Ethical

document

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cause

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hacko's ment

minimize

harm,

incidents immediately & take

responsibility

report

for their

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Testin

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Without

Ethical

1) the

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2) Emerge ney

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IT Department

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